# Soil Ring Shear Sensor Calibration

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Soil ring shear instrumentation consists of: 3 load cells, 4 LVDTs and 1 torque sensor.

## Load cell calibration

Load cells were calibrated using the GCTS hydraulic frame (Weeks Hall, B138, Figure 1) and Transducer Techniques calibrated load cell (Figure 2).

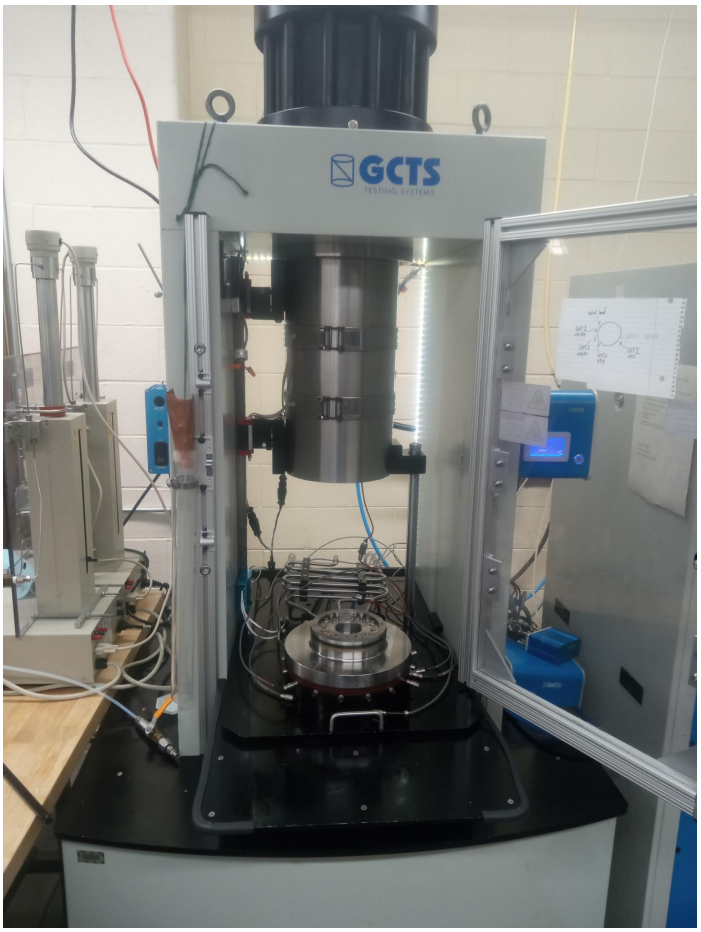


Figure 1: GCTS hydraulic load frame.

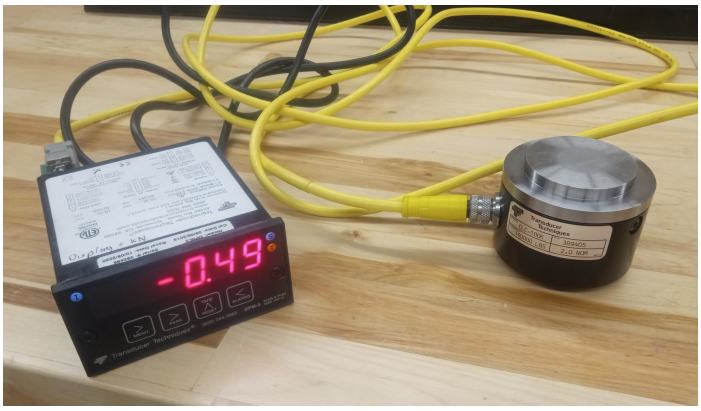


Figure 2: Transducer Techniques calibrated load cell.

The displacement of the GCTS hydraulic piston was controlled and both the Transducer Techniques and the soil ring shear load cells were compressed with the same force. The force reading from the TT load cell was taken at several points during loading. The calibration results are presented in Figure 3.

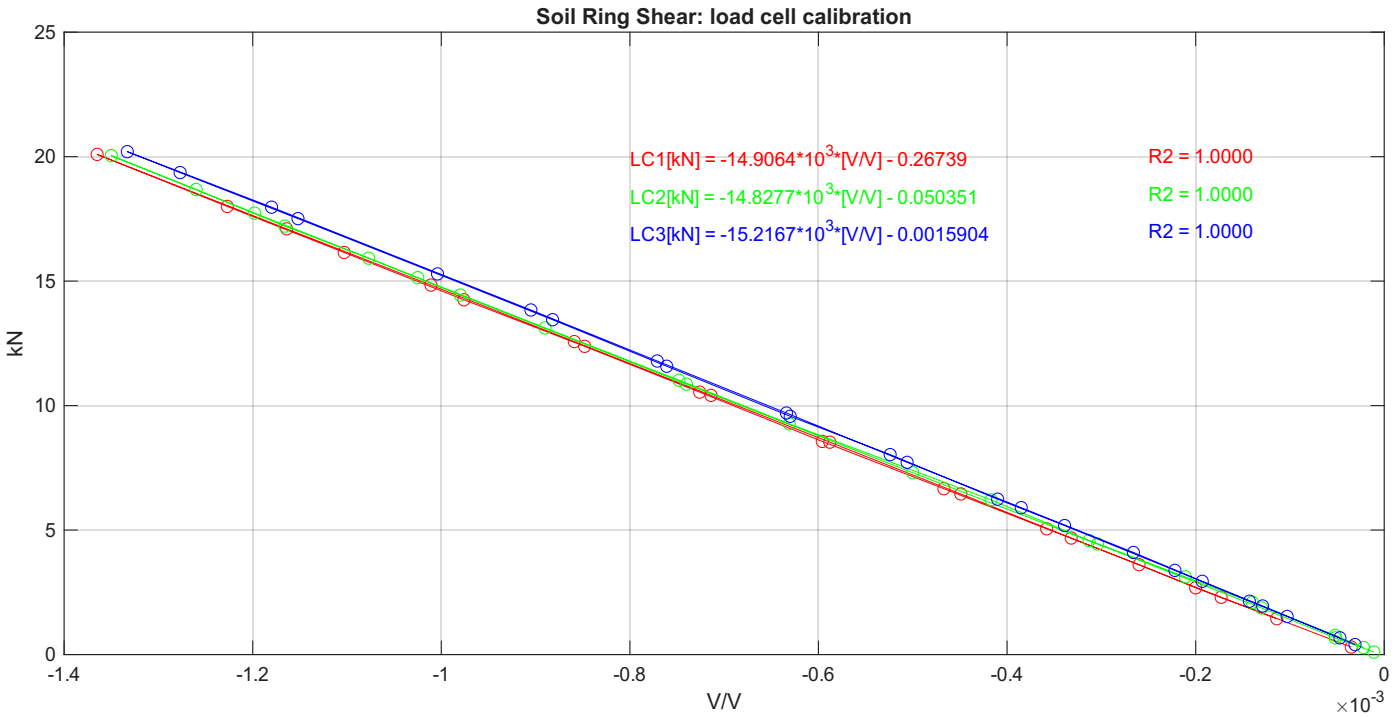


Figure 3: Soil ring shear load cell calibration results.

# LVDT calibration

LVDTs were calibrated using the LVDT calibration device (Figure 4).

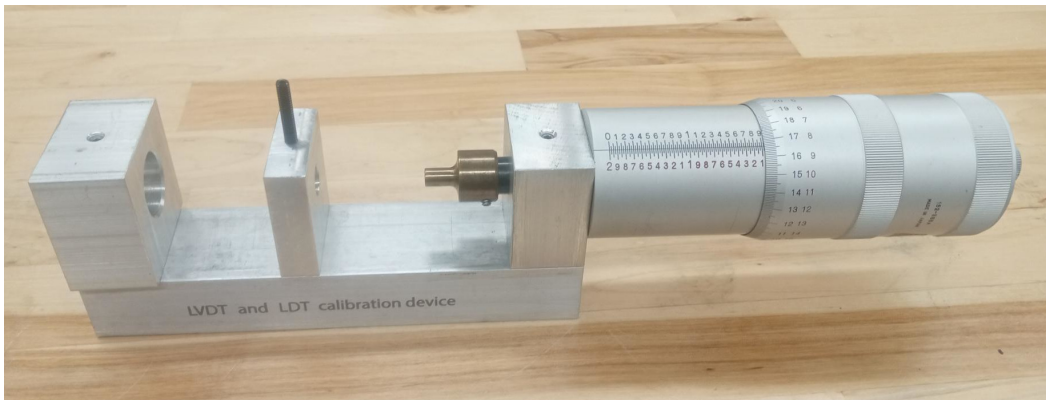


Figure 4: LVDT calibration device.

The LVDT under calibration is secured in the appropriate size adapter on the left, and the dial is used to change the extension of the LVDT rod. The dial reads inches. The calibration results are presented in Figure 5.

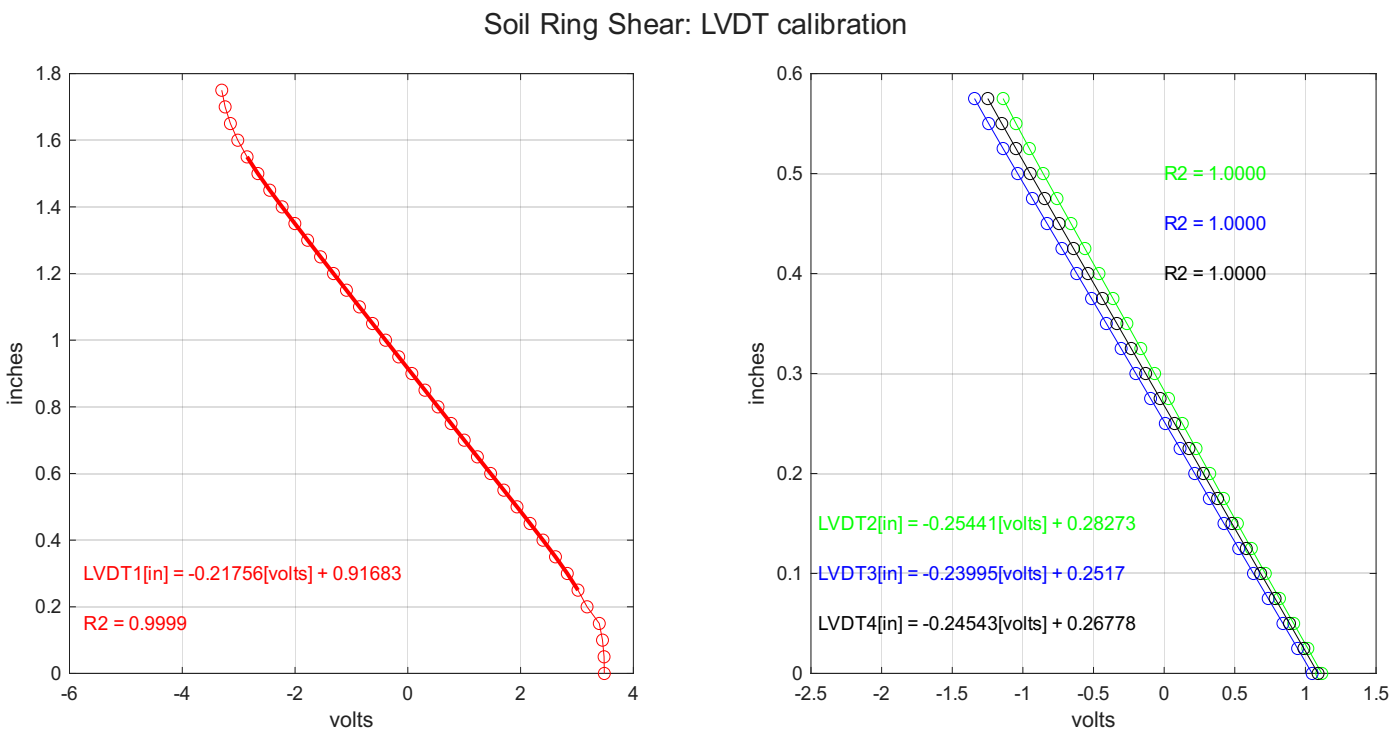


Figure 5: Soil ring shear LVDT calibration results.

## Torque sensor calibration

The torque sensor calibration setup is presented in Figure 6.

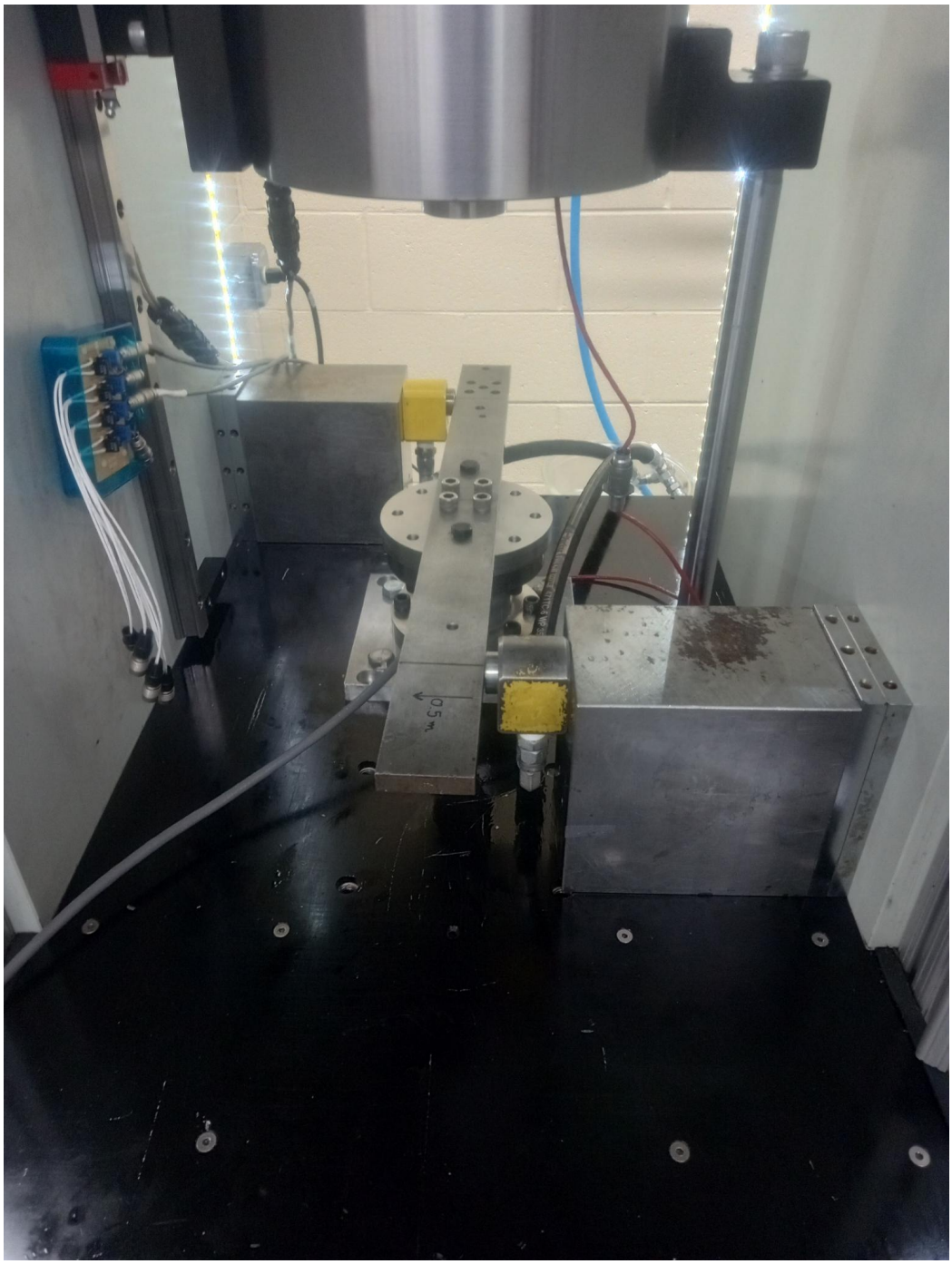

Figure 6: Torque sensor calibration setup.

The torque sensor was calibrated in the following steps:

- the sensor was bolted to the GCTS hydraulic frame
- a stainless steel disk with a steel bar was attached to the top of the torque sensor
- hydraulic pistons (yellow) were installed at $r$=8.375" distance from the center of the sensor on both sides
- pressure was applied to the hydraulic pistons which converted to known pushing force on the steel bar

Using the GCTS pressure intensifier, we were controlling the pressure supplied to the hydraulic pistons. Then, the torque was calculated from:

$$T = 2 \cdot P \cdot A_{piston} \cdot r \tag{1}$$

where $P$ is the applied pressure, $A_{piston}$ is the area of the piston (1 sq. in.), $r$ is the distance (8.375”). The factor of 2 is there because we are using two pistons on opposite sides with the same arm length and pressure. With this setup it is possible to also apply normal load to the torque sensor using the GCTS hydraulic frame piston. The current calibration was performed without additional normal stress.

The calibration results are presented in Figure 7.

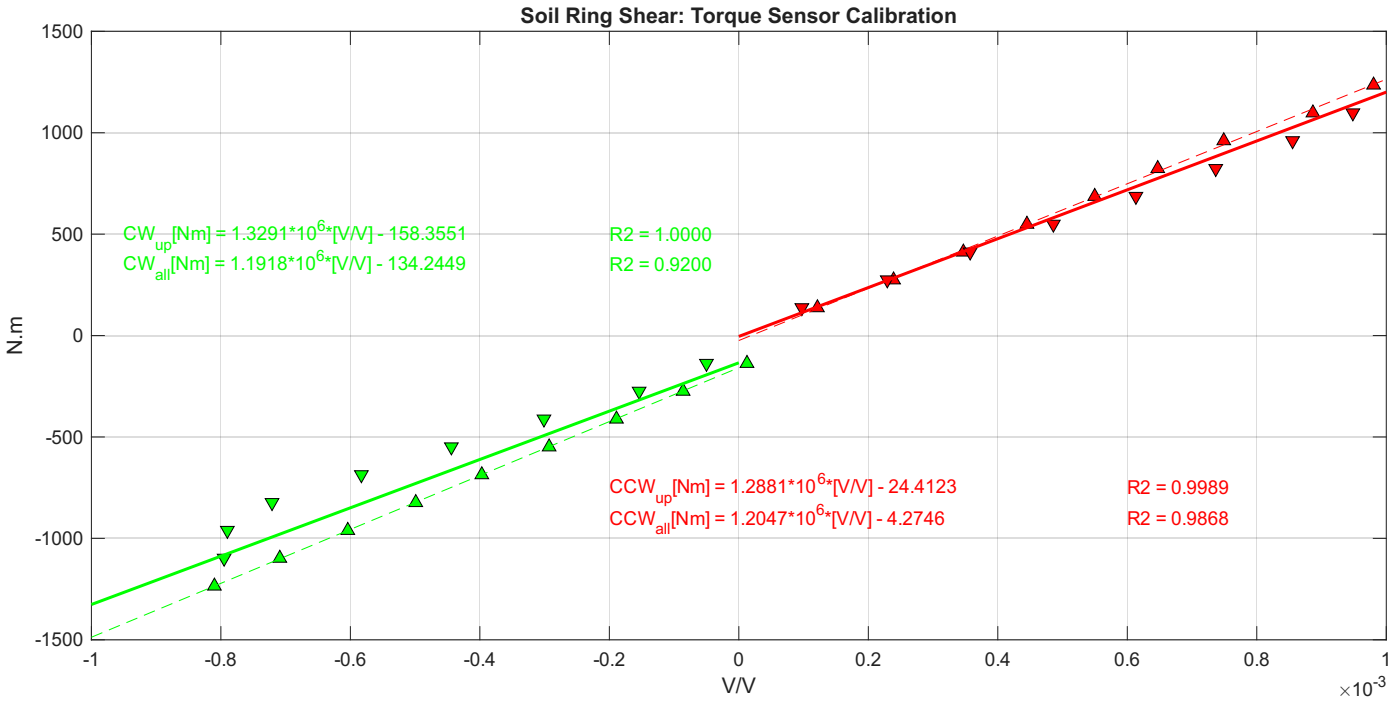


Figure 7: Soil ring shear torque sensor calibration results.